JavaScript Functions — Night Revision Sheet

Sigma Web Development Course | Quick Reference & Code Examples

1. Function Definition & Reuse (Work Concept)

What is a Function? Function ek reusable code block hota hai jo specific work (kaam) perform karta hai. Yeh code repetition ko khatam karta hai aur programming ko structured banata hai.

```
// Definition: Creating a function to do repetitive work
function nice(name) {
    console.log("Hey " + name + " you are nice");
    console.log("Hey " + name + " your course is good too");
}

// Function Invocation (Calling the function)
nice("Harry");
nice("Rohan");
```

2. Return Values & Parameters

Functions inputs (parameters) le sakte hain aur `return` keyword ki madad se calculated result ko caller ko wapas bhej sakte hain.

```
// Function performing addition work and returning result
function sum(a, b) {
    return a + b; // Sends output back
}

let result1 = sum(3, 5);  // Output: 8
let result2 = sum(7, 5);  // Output: 12
console.log("Sum is:", result1);
```

Quick Revision Points:

- **Parameter:** Function definition ke time par jo variables diye jaate hain (e.g., a , b).
- **Argument:** Function call karte waqt pass kiye jaane waale actual values (e.g., 3 , 5).
- **Return:** Function se output vaapis lene ke liye use hota hai.

Advanced Function Concepts

3. Default Parameters

Agar function call karte waqt koi argument missing ho, toh JavaScript default value ko use karti hai.

```
// 'c' has a default value of 3
function sumWithDefault(a, b, c = 3) {
    return a + b + c;
}

// Only 2 arguments passed, c takes default value 3
console.log("Result:", sumWithDefault(5, 3)); // 5 + 3 + 3 = 11
```

4. Arrow Functions

Modern JavaScript (ES6) ka concise syntax jismein functions ko variables ke andar store kiya jata hai.

```
// Arrow function doing work and stored in variable 'func1'
const func1 = (x) => {
    console.log("Arrow function doing work with input:", x);
}

func1(34);
func1(66);
```

Night Revision Checklist:

- **DRY Principle:** "Don't Repeat Yourself" — Functions code ko repeat hone se bachate hain.
- **Arrow Functions:** Short clean syntax, modern JS frameworks (React, Node) mein bohot use hota hai.
- **Default Values:** Function parameters mein unexpected `undefined` errors se bachati hain.